Sixteen integer unknowns for an explicit consistency equation

and simpler polynomial presentations of Con(ZFC)

Research synthesis for the ProveIt project

Prepared with AI assistance; ordinary mathematical proofs and reproducible artifacts

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Abstract

We combine the current ProveIt constructions with a positivity-normalization lemma for Pell equations. The resulting explicitly specified integer polynomial P_{16} has sixteen unrestricted integer unknowns and satisfies

$$\mathsf{ZFC} \vdash \left(\nexists \mathbf{x} \in \mathbb{Z}^{16} : P_{16}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

Its degree is exactly $406528 \cdot 5^{32} + 10008$ and its delivered arithmetic circuit uses 1590 binary operations. The reduction removes a guard witness from the existing seventeen-integer construction: a signed Pell coordinate can be replaced by an arbitrarily large positive coordinate in the same residue class. A parity-sensitive radical weight lowers the degree further. We also obtain a 61-operation system of twenty-two quadratics in 31 positive unknowns, with a 126-operation quartic sum of squares, and an identical 108-operation evaluator for the existing degree-1081 polynomial in twelve positive unknowns. A separate complete native coefficient tuple reduces the radix from the previous best $2^{1,953,125}$ to $2^{1,012,625}$. The arithmetic circuits, finite coefficient recipes, exact algebra checks, source pins, and reproduction scripts are included. These are improvements to explicit presentations in the inspected repository, not global minimality claims or a proof of consistency. The new proofs are ordinary mathematics; the inherited native compiler and Jones representation theorems remain identified dependencies.

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1 What is improved, and what "simplest" means

The reviewed repository revision is `8c6eec9d19ef80538c8e61ff6e0180040583446a`. The target directory is `con-zfc-diophantine-forms`, under `SetTheory/Cardinals/docs/reports/hilbert-tenth-problem`; its audited README describes seven wave-4 packages, numbered 24–30 [1]. The source snapshot is retained in the archive. Different presentations optimize different quantities; number of variables, degree, coefficient height, and arithmetic evaluation cost cannot be collapsed into one ordering.

Measure	Inspected predecessor	This synthesis
Explicit unrestricted-integer construction	17 variables; 20,607 operations	16 variables; 1590 operations
Degree of that construction	$424296 \cdot 5^{33} + 53824$	$406528 \cdot 5^{32} + 10008$
31-positive-variable quadratic system	22 equations; 65 operations	22 equations; 61 operations
31-positive-variable quartic	130 operations; 306 symbolic monomials	126 operations; 259 symbolic monomials
12-positive-variable degree-1081 polynomial	111 operations	108 operations; exactly the same polynomial
Associated three-equation system	107 operations	104, or 103 after a justified coordinate shift
Native coefficient radix	$2^{1,953,125}$	$2^{1,012,625}$; separate complete tuple

All operation counts charge binary addition, subtraction, and multiplication, including scalar multiplication and squaring. Sharing an already computed value and comparing two system sides are free. Fixed integer coefficients are leaves, not additional unknowns. Fixed exponentiation is expanded into multiplications. Thus these are arithmetic-circuit sizes, not bit-complexity bounds and not measurements of the work needed to construct the coefficients. The monomial comparison keeps the four native coefficient ports symbolic.

The repository's nine-natural-variable constructions remain better in variable count over $\mathbb{N}$; its 67-operation single equation remains better in operation count, and its 59-operation systems remain better in that metric. We do not claim a smaller universal variable bound. In particular, Sun's 2026 preprint gives an effective general representation theorem in ten unrestricted integers [7]; that different compiler is not instantiated here. Our sixteen-variable claim concerns the concrete, fully bound constructions supplied with this report.

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$. An equation over $\mathbb{Z}_{>0}$ must not silently be read over $\mathbb{Z}$. Our main P_{16} does allow every one of its sixteen variables to range over all of $\mathbb{Z}$.

2 Fixed source instance and logical dependencies

The logical statement Con(ZFC) means that no finite proof of contradiction exists in the fixed recursive presentation of ZFC used by the repository. The calibrated machine is the literal 215-state machine `candidate-k215/k215.tm`. Its Lean endpoint states

```
blank_halts_iff_zfc_refutation :
  BlankHalts machine215 <-> BProv ZFCax_s [] Form.fBot
```

The source file is included. This report neither rebuilds that development nor relies on a metatheoretic assertion that Lean is sound to infer a theorem of ZFC. We use the ordinary arithmetic calibration mirrored by the endpoint, then compose ordinary mathematical reductions formalizable in ZFC.

The native compiler contributes four fixed integers

$$\beta = B_0 - 1, \qquad K_0 = D_C + B_0 D_R, \qquad M_C, \qquad M_F = M_F^{\text{native}} + B_0 - 1, \qquad B_0 = 2^{5^{10}}.$$

Throughout the main construction these are exactly the complete compact943 tuple of package 30. In particular M_F is the *shifted* mask. The four gzip payloads are included, together with the original binding, layout, ordered 943-tile list, and coefficient proofs. The raw sizes are

Integer	Binary digits	Raw bytes
β	9,765,625	1,220,704
K_0	13,201,251	1,650,157
M_C	9,765,625	1,220,704
M_F	9,765,626	1,220,704

The total is 5,312,269 raw bytes, stored in 20,260 gzip bytes. Byte hashes are checked during reproduction. Compression does not change coefficient height. No coefficient is chosen by testing whether ZFC is consistent.

The imported compiler theorem identifies positive solutions of the parent quartic with blank halting of K215. Its hypotheses include the native-window language, no-carry bounds, synchronization, rank control, and the Pell arithmetization; they are not assertions about arbitrary choices of four integer ports. The source proofs are bundled, and the new transformations preserve this one coherent tuple. The Jones reduction is the concrete quartic-to-ratio construction of [6], mirrored by `Jones1982.Nine.ratio_sound`, `ratio_complete`, and `wset_iff_nine` in the repository.

3 A cheaper positive quartic

All thirty-one variables in this section are strictly positive:

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, Y, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

Use only the affine abbreviations

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha, \quad E = a - Y.$$

The new system consists of the following twenty-two quadratic equations:

$$\begin{aligned}
L &= q^2 - 1, & N &= qL,\\
X &= wq, & Y &= s(N + q),\\
E &= XY, & c &= kY + \eta,\\
P &= E(c - \eta), & \tau^2 &= P(P + k) + 1,\\
D &= X + ac + \gamma H, & \Delta &= a^2 + H,\\
C_2 &= c^2, & D^2 &= \Delta C_2 + 1,\\
k &= jc - V + 1 + hE,\\
jc - V &= (q - F)N - CL + (M_C + qM_F)J,\\
(K_0 + w)C &= F + z(q - 1),\\
F_2 &= f^2, & V &= of - c,\\
g &= y^2 - V^2, & I &= iC_2,\\
B_{\mathrm{aux}} &= I^2, & B_{\mathrm{aux}} &= \Delta(F_2 - 1),\\
y^2 - B_{\mathrm{aux}} g &= 1. & &
\end{aligned} \tag{1}$$

Let $r_1, \ldots, r_{22}$ be these left sides minus right sides, in row order, and define $Q^*_{31} = \sum r_j^2$. The JSON gives this order unambiguously.

Proposition 3.1. *The positive zero set of* (1) *is in bijection with the positive zero set of package 30's parent quartic. Consequently*

$$\mathsf{ZFC} \vdash (\nexists \mathbf{x} \in \mathbb{Z}_{>0}^{31} : Q_{31}^{*}(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The degree is exactly four. The system has a 61-operation circuit, and the displayed sum of squares has a 126-operation circuit.

Proof. Write a subscript o for parent coordinates. Apply the simultaneous affine change

$$N_o = N + q, \qquad E_o = a - Y, \qquad P_o = P + k, \qquad z_o = z + 1,$$

leaving all other variables fixed. Every parent residual becomes exactly the corresponding residual of (1). All twenty-two identities are checked as sparse integer-polynomial identities with symbolic coefficient ports. The inverse change is $N = N_o - q$, $Y = a - E_o$, $P = P_o - k$, $z = z_o - 1$.

A positive new zero restores $E_o = XY > 0$; the other three restored coordinates are visibly positive. Conversely the parent equations give $N = qL > 0$, $Y = sN_o > 0$, and $P = E_o kY > 0$. For the fourth coordinate we use the parent's established transport-quotient lemma, also used in its retained59 construction. After the Pell/compiler reconstruction it gives $C > 0$, $X = wq = 2^R$, $R > 3q$, and $F < q$. Thus $w > F$, and

$$0 < (K_0 + w)C - F = (z_o - 1)(q - 1)$$

forces $z_o \geq 2$. This is the necessary positive-domain argument; positivity of z_o alone would not suffice.

Each of the four changes saves one gate in the scheduled system evaluation. For example $P_o = E_o(c - \eta) + k$ loses its final addition, while its norm replaces $P_o - k$ by $P + k$ at the same cost. The N change deletes two operations and adds the one operation in $N + q$. The result is $61 = 33$ multiplications $+28$ additions/subtractions. Forming twenty-two residuals, twenty-two squares, and their sum costs $22 + 22 + 21 = 65$ more. Finally τ appears only as τ^2 in r_8; the coefficient of τ^4 in the sum of squares is exactly one. □

4 A finite, fully specified Jones index

In this and later sections all letters are newly scoped. Put $\nu = 31$ and $\mathcal{L} = 5^{32}$. Define

$$Q(x, n_1, \ldots, n_{31}) = Q_{31}^{*}(n_1 + 1, \ldots, n_{31} + 1),$$

independently of the dummy input x. At the all-zero natural tuple the fifth residual is -1, so $Q(x, 0) > 0$. Thus Q satisfies the normalization hypothesis of the concrete Jones theorem, and has degree four.

For $\alpha \in \mathbb{N}^{32}$ with $|\alpha| \leq 4$, let p_α be its ordinary coefficient and put

$$c_\alpha = (4 - |\alpha|)! \prod_{j=0}^{31} \alpha_j! \, p_\alpha, \qquad e(\alpha) = \sum_{j=0}^{31} \alpha_j 5^j.$$

There are 391 nonzero symbolic records, each an explicitly listed polynomial in (β, K_0, M_C, M_F); the shifted quartic has 571 terms with these ports symbolic. The new coefficient file is `data/coefficients31.json`. It is freshly derived from Q_{31}^{*}, not obtained by reusing the parent's old coefficient list after changing the parent.

For a record $c_\alpha = \sum_{\mathbf{e}} a_{\mathbf{e}} \prod_{j=1}^{4} b_j^{e_j}$, use the authentic port bit lengths ℓ_j in Section 2 and the finite integer bound

$$|c_\alpha| \leq H_\alpha := \sum_{\mathbf{e}} |a_{\mathbf{e}}| 2^{\sum_j e_j \ell_j}.$$

The supplied exact arithmetic certificate verifies $H_\alpha + 1 < 2^{45933757}$ for every record. Unlisted coefficients are zero. Set, once and for all,

$$\begin{aligned} z_* &= 2^{45933757}, \\ e_*(T) &= \sum_{|\alpha|\le 4} (z_* + c_\alpha) T^{\mathcal{L}-e(\alpha)}, \\ y_* &= e_*(2z_*), \qquad u_* = \sum_{j=1}^{31} (2z_*)^{5^j}. \end{aligned}$$

All exponents are fixed nonnegative integers; in particular $e(\alpha) \le 4 \cdot 5^{31} < \mathcal{L}$. The sum has $\binom{36}{4} = 58905$ terms. Its base-$2z_*$ digits are strictly between 1 and $2z_*$, in distinct positions. These definitions satisfy precisely the admissible index hypothesis for the normalized quartic Q. The unused u_* is specified to make that interface explicit.

This is a finite definition of literal integer coefficients. Materializing the enormous integer y_* is unnecessary to specify the polynomial. It is not an unspecified recursive-enumeration index, an existential parameter, or a value defined by whether a computation halts.

5 The eight-coordinate ratio predicate

Use the eight base coordinates

$$(\epsilon, g, h, s, w, \phi, i, j).$$

Initially, Jones's predicate uses $\epsilon, \phi \ge 0$ and $g, h, s, w, i, j \ge 1$. Later h, i, j will range over all integers. All other symbols in this section are polynomial abbreviations:

$$b = y_* + 1 + \epsilon, \quad B = b^5, \quad q = B^{\mathcal{L}}, \quad \ell = \sum_{r=1}^{31} B^{5^r}, \quad \lambda = \sum_{r=0}^{4\mathcal{L}-1} B^r,$$
$$e = e_*(B), \quad \theta = B - 2z_*, \quad N = q^{16}, \quad G_c = 32bB^{5^{31}}.$$

Define R by

$$\begin{aligned} R = {} & [g + eq^3 + \ell q^5 + \{2(e - z_*\lambda)(1 + B + g)^4 + \lambda B + \lambda B q^4\} q^7](N^2 - N) \\ & + [q^3 - b\ell + \ell + \theta\lambda q^3 + (B-2)q^8](N^2 - 1). \end{aligned}$$

The remaining abbreviations are

$$\begin{aligned} U &= N^2 w, & Y &= N^2 s, & M &= RY, \\ P &= 2M^2 U, & K &= R + 1 + h(P-1), & A &= M(U+1), \\ B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi, \\ J &= 4A - 5, & Q_1 &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2-1)C^2 + 1, \\ E &= iJDC^2, & F &= (A^2-1)E^2 + 1, & G_1 &= A + F(CD + 1 - A), \\ H &= B_1 + jC, & I &= (G_1^2 - 1)H^2 + 1. \end{aligned}$$

Finally put

$$\begin{aligned} a_0 &= (P^2-1)K^2 + 1, \qquad a_1 = DFI, \\ m &= K^2 G_c - 4(C - KY)^2 G_c - 8gK^2, \\ D_0 &= JF, \qquad T_0 = Q_1 F + J(H - C). \end{aligned}$$

The imported concrete Jones equivalence says

$$\begin{aligned}&\exists \text{ base coordinates in the initial domain}: \quad a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}),\ D_0 \mid T_0,\ m > 0\\ &\qquad\Longleftrightarrow \quad \exists \mathbf{n} \in \mathbb{N}^{31} : Q(1, \mathbf{n}) = 0.\end{aligned} \tag{JR}$$

This is the ratio predicate underlying the nine-variable theorem, not a claim that an arbitrary nine-variable polynomial has this structure.

6 Removing three positivity restrictions

Lemma 6.1. *For every integer assignment of the eight base coordinates, $D_0 \neq 0$. Replacing a nonzero i by $|i|$ preserves a_0, a_1, D_0, T_0, m.*

Proof. $J = 4A - 5 \neq 0$. If $F = 0$, then $(1 - A^2)E^2 = 1$, forcing $A = 0$ and $E = \pm 1$. But then $J = -5$ and $E = iJDC^2$ is divisible by 5, a contradiction. The only nonsquared occurrence of i is in E, which enters the five listed quantities only through E^2. □

Lemma 6.2. *Suppose $\epsilon, \phi \geq 0$, $g, s, w \geq 1$, $a_0 \in \mathrm{Sq}(\mathbb{Z})$, and $m > 0$. Then $h \geq 1$. In particular the proof does not assume a sign for j.*

Proof. The index gives $b \geq 2$, $B = b^5 > 2z_*$, and $e > 0$. Since $G_c > 0$ and $g > 0$, the margin implies $K \neq 0$,

$$g < G_c/8 = 4bB^{5^{31}}, \qquad 4(C - KY)^2 < K^2.$$

Writing $t = 5^{31}$, we have $1 + B + g < 1 + B + 4B^{t+1} \leq 5B^{t+1} < B^{t+2} \leq B^{5t} = q$. Also $\ell < B^{t+1}$, hence $b\ell < B^{t+2} \leq q$. In the left bracket for R, the possibly negative term $-2z_*\lambda(1 + B + g)^4$ is dominated by $\lambda B q^4$. Every remaining term there is nonnegative and g is positive. The right bracket is positive as well. Thus $R \geq N^2 - N \geq 2$.

Now $Y, U \geq 1$, $C = 2R + 1 + bw + \phi > 0$, and $P = 2R^2Y^2U \geq 2R^2 > R + 2$. The margin excludes $K \leq 0$: if $K < 0$, then $|C - KY| > |K|$, and $K = 0$ was already excluded. If $h \leq 0$, then $1 \leq K \leq R + 1 < P$. The case $K = 1$ would give $P - 1 \mid R$, impossible because $0 < R < P - 1$. For $2 \leq K < P$,

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2,$$

as the two differences are $K(2P - K) > 0$ and $K^2 - 1 > 0$. This contradicts squarehood of a_0. □

The next lemma supplies the new witness saving.

Lemma 6.3 (Positive Pell coordinates in a prescribed residue class)**.** *Let $a \geq 2$ be an integer and $d = a^2 - 1$. If $x^2 - dy^2 = 1$ with $x > 0$ and $y \in \mathbb{Z}$, then for every modulus $M \geq 1$ there are solutions (x', y') with $x', y' > 0$, arbitrarily large y', and $x' \equiv x$, $y' \equiv y \pmod{M}$.*

Proof. The element $u = a + \sqrt{d}$ is a unit of norm one in $\mathbb{Z}[\sqrt{d}]$ and its multiplication matrix has determinant one. In the finite group $\mathrm{SL}_2(\mathbb{Z}/M\mathbb{Z})$ some positive power u^r is the identity. Multiply $x + y\sqrt{d}$ by u^{rn}. Its coordinates are integers, its norm stays one, and both coordinates retain their residues modulo M. The real number $x + y\sqrt{d}$ is positive: its conjugate has the same sign and their sum is $2x > 0$. Since $u > 1$, the second coordinate is

$$\frac{(x + y\sqrt{d})u^{rn} - (x + y\sqrt{d})^{-1}u^{-rn}}{2\sqrt{d}} \longrightarrow +\infty.$$

The first coordinate is positive as well. This proves the claim, including negative or zero starting second coordinates. □

Proposition 6.4 (The j guard is unnecessary)**.** *The existential predicate* (JR) *is unchanged if its base domain is replaced by*

$$\epsilon, \phi \geq 0, \qquad g, s, w \geq 1, \qquad h, j \in \mathbb{Z}, \qquad i \in \mathbb{Z} \setminus \{0\}.$$

Proof. Only soundness of the larger domain needs proof. Lemmas 6.1 and 6.2 normalize i and show $h \geq 1$. We now have $A > 1$, $C > B_1 > 0$, and $G_1 > 1$. The congruences $F \equiv 1 \pmod{D}$ and $F \equiv 1 \pmod{J}$ follow from $E = iJDC^2$. Consequently $D_0 \mid T_0$ implies $F \mid H - C$. Modulo D we have $G_1 \equiv 1$, hence $I \equiv 1$. Modulo F, $G_1 \equiv A$ and $H \equiv C$, hence $I \equiv D$. Therefore D, F, I are pairwise coprime positive integers. Squarehood of DFI forces each of them to be a square.

Choose the positive integer $x = \sqrt{I}$; then $x^2 - (G_1^2 - 1)H^2 = 1$. Apply Lemma 6.3 with $a = G_1$ and modulus CF. Choose $H' > B_1$ congruent to H modulo CF, and set $j' = (H' - B_1)/C$. This is a positive integer, since $H \equiv B_1 \pmod{C}$. The new I' is a square, so DFI' is a square. Also

$$T_0' - T_0 = J(H' - H) \quad \text{is divisible by } JF.$$

All other ratio conditions remain unchanged. We have recovered a tuple in the original domain, to which (JR) applies. The map is existential: it changes the numerical value of I, while preserving its required squarehood and the required congruences. □

7 Encoding the remaining guards

Introduce six unrestricted integer variables $z_1, \ldots, z_6$ and put

$$(d_1, d_2, d_3, d_4, d_5, d_6) = (3\epsilon + 2, 3g - 1, 3s - 1, 3w - 1, 3\phi + 2, 3m - 1).$$

Starting with $C_0 = 1$, define

$$B_r = 1 + 4d_r C_{r-1}^2 z_r^2, \qquad C_r = C_{r-1} B_r \quad (1 \leq r \leq 6), \qquad C_* = C_6.$$

Lemma 7.1. *Subject to $z_1 \cdots z_6 \neq 0$, the condition $C_* \in \mathrm{Sq}(\mathbb{Z})$ is equivalent, existentially in the six z_r, to $\epsilon, \phi \geq 0$, $g, s, w \geq 1$, and $m > 0$. Moreover $C_* \equiv 1 \pmod 4$ for every integer assignment, without any squarehood or nonvanishing hypothesis.*

Proof. Each $B_r \equiv 1 \pmod 4$ is nonzero, and $\gcd(B_r, C_{r-1}) = 1$. Thus the B_r are pairwise coprime. If their product is a square, each $|B_r|$ is a square; a negative square is 0 or 3 modulo 4, so each B_r is a positive square. Because $C_{r-1} z_r \neq 0$, this forces $d_r \geq 0$. Its residue 2 modulo 3 excludes zero, giving exactly the six desired inequalities.

Conversely each positive $d_r \equiv 2 \pmod 3$ is nonsquare, so its Pell equation $u^2 - d_r v^2 = 1$ has a nontrivial positive solution. A power of its unit is the identity modulo $2|C_{r-1}|$, by the same finite-group argument as above. We may therefore take v a positive multiple of $2|C_{r-1}|$ and set $z_r = v/(2C_{r-1}) \neq 0$. Then $B_r = u^2$. Induction proves the converse. The final congruence follows directly from the formulas. □

8 One divisor for divisibility and nonvanishing

The existing signed construction uses the degree-five intersective polynomial

$$\Pi(t) = (t^3 - t - 1)(t^2 + 23).$$

For completeness the necessary arithmetic is proved here.

Lemma 8.1. Π *has no rational root but has a root modulo every nonzero integer.*

Proof. The monic cubic has only ± 1 as possible rational roots, and neither works. The quadratic has no real root. The cubic discriminant is -23. For an odd prime $p \neq 23$, a cubic root modulo p is simple and lifts. If there is no cubic root, the cubic is irreducible. Frobenius permutes its three roots by a 3-cycle, so their Vandermonde product is fixed; its square is -23. The quadratic consequently has a simple root and lifts instead. At $p = 23$, the cubic root 3 is simple because its derivative is 26. At $p = 2$, use $-23 \equiv 1 \pmod 8$. A root modulo 2^e for $e \geq 3$ lifts to 2^{e+1} by choosing one of t and $t + 2^{e-1}$. Roots modulo arbitrary integers now follow by the Chinese remainder theorem. □

Define the polynomial

$$\mathcal{R}(D, T, v) = \big((T + Dv)^3 - D^2(T + Dv) - D^3\big)\big((T + Dv)^2 + 23D^2\big).$$

Lemma 8.2. *For integers D, T, Z with $D \neq 0$,*

$$(D \mid T \ \wedge \ Z \neq 0) \quad \Longleftrightarrow \quad \exists v \in \mathbb{Z} : \ ZD^5 \mid \mathcal{R}(D, T, v).$$

Also $\mathcal{R}(D, T, v) \neq 0$ for every integer v.

Proof. The identity $\mathcal{R} = D^5\Pi(T/D + v)$ proves nonvanishing. If $T = Dq$ and $Z \neq 0$, choose v by Lemma 8.1 so that $Z \mid \Pi(q + v)$. Conversely $Z = 0$ would force $\mathcal{R} = 0$. If $D \nmid T$, choose a prime with $\alpha = v_p(D) > \beta = v_p(T)$. The first term in each factor of $\mathcal{R}$ has uniquely least valuation: respectively 3β and 2β. Thus $v_p(\mathcal{R}) = 5\beta < 5\alpha$, contradicting divisibility by D^5. □

9 The polynomial in sixteen unrestricted integers

The sixteen variables are exactly

$$\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, z_4, z_5, z_6, v, u \in \mathbb{Z}.$$

Using Sections 5–7, put

$$Z = iz_1z_2z_3z_4z_5z_6, \qquad S = ZD_0^5, \qquad T = \mathcal{R}(D_0, T_0, v), \qquad X = Su - T,$$
$$w_1 = C_* + 1, \qquad w_2 = w_1(P^2 + K^2 + 2),$$
$$A_0 = S^2C_*, \qquad A_1 = (Sw_1)^2a_0, \qquad A_2 = (Sw_2)^2a_1,$$
$$V = X^2 + A_2 - A_0 - A_1, \qquad W = V^2 + 4X^2A_2 - 4A_0A_1.$$

Our equation is the ordinary integer-polynomial equation

$$\boxed{P_{16} = W^2 - 16V^2X^2A_2 = 0.} \tag{2}$$

Every intermediate letter is an abbreviation. In particular none adds a quantifier. The coefficient leaves are completely fixed in Sections 2 and 4.

Lemma 9.1 (Parity-sensitive separation)**.** *At every integer assignment,*

$$|w_1| > \sqrt{|C_*|}, \qquad |w_2| > \sqrt{|C_*|} + |w_1|\sqrt{|a_0|}.$$

Proof. Because $C_* \equiv 1 \pmod 4$, either $C_* \geq 1$ or $C_* \leq -3$. In both cases $(C_* + 1)^2 > |C_*|$. For arbitrary real P, K,

$$\sqrt{|(P^2 - 1)K^2 + 1|} \leq P^2 + K^2 + 1.$$

Indeed bound the absolute radicand by $(P^2 + 1)K^2 + 1$ and square the nonnegative right side; the remaining difference is $P^4 + K^4 + P^2K^2 + 2P^2 + K^2 \geq 0$. Multiplying by $|w_1|$ proves the second strict inequality. Negative values of w_1, w_2 cause no issue. □

Lemma 9.2 (Three-root finalizer)**.** *For fixed earlier coordinates,* (2) *has an integer solution in* u *if and only if*

$$S \neq 0, \qquad S \mid T, \qquad C_*, a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}).$$

Proof. Lemmas 6.1 and 8.2 give $T \neq 0$. If $S = 0$, then $P_{16} = T^8 \neq 0$. Assume $S \neq 0$, and choose complex roots $r_0^2 = C_*$, $r_1^2 = a_0$, $r_2^2 = a_1$. Exact multiplication gives

$$P_{16} = \prod_{\delta_0,\delta_1,\delta_2 \in \{\pm 1\}} [X - S(\delta_0 r_0 + \delta_1 w_1 r_1 + \delta_2 w_2 r_2)].$$

A zero makes one signed radical sum equal to the rational number $u - T/S$. Two distinct sign patterns on nonzero roots cannot give the same sum: take their largest differing index; a nonzero square root of an integer has absolute value at least one, and Lemma 9.1 makes its weighted contribution exceed the sum of all preceding contributions. The triangle inequality proves this even for negative radicands.

The field generated by the three roots is a finite Galois extension of $\mathbb{Q}$, and its automorphisms change their signs. Every automorphism fixes the rational sum; separation therefore forces it to fix each root individually. Each root lies in $\mathbb{Q}$, hence in $\mathbb{Z}$ because its square is an integer. Zero roots are already rational. Thus all three radicands are nonnegative integer squares and T/S is an integer. Conversely, choose their integer roots and set $u = T/S + r_0 + w_1 r_1 + w_2 r_2$. □

Theorem 9.3 (Main equivalence)**.** *For the fixed coefficients defined above,*

$$\mathsf{ZFC} \vdash (\nexists \mathbf{x} \in \mathbb{Z}^{16} : P_{16}(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

Proof. A zero gives the conditions of Lemma 9.2. Since $D_0 \neq 0$, $S \neq 0$ implies $iz_1 \cdots z_6 \neq 0$. Lemma 8.2 recovers $D_0 \mid T_0$, and Lemma 7.1 recovers $\epsilon, \phi \geq 0$, $g, s, w \geq 1$, $m > 0$. Proposition 6.4 restores $h, i, j > 0$ existentially. The concrete Jones equivalence, the quartic equivalence, and the K215 calibration now give a finite ZFC refutation.

Conversely a refutation supplies a positive quartic solution and then a base tuple satisfying (JR). Lemma 7.1 supplies six nonzero guard witnesses with $C_* \in \mathrm{Sq}(\mathbb{Z})$. Lemma 8.2 supplies v with $S \mid T$; here $S \neq 0$. Lemma 9.2 supplies u. Negating the existence equivalence proves the claim. The steps use integer arithmetic, finite-group arguments, elementary Pell theory, and finite Galois theory, all formalizable in ZFC. □

10 Exact degree and circuit specification

Write $\mathcal{L} = 5^{32}$. Degrees below treat every fixed coefficient as degree zero. The relevant source degrees are

Quantity	Degree	Quantity	Degree
R	$235\mathcal{L}$	P	$950\mathcal{L} + 3$
K	$950\mathcal{L} + 4$	a_0	$3800\mathcal{L} + 14$
a_1	$24640\mathcal{L} + 68$	m	$2221\mathcal{L} + 11$
D_0	$6875\mathcal{L} + 20$	C_5	363
C_*	$2221\mathcal{L} + 1102$	w_1	$2221\mathcal{L} + 1102$
w_2	$4121\mathcal{L} + 1110$	S	$34375\mathcal{L} + 107$
T	$34375\mathcal{L} + 105$	X	$34375\mathcal{L} + 108$
A_2	$101632\mathcal{L} + 2502$	P_{16}	$406528\mathcal{L} + 10008$

The first five stacks obey $c_r = 3c_{r-1} + 3$, $c_0 = 0$; the final stack uses the margin. A_2 has strictly higher degree than X^2, A_0, A_1. Thus the leading terms of V, W, P_{16} are respectively A_2, A_2^2, A_2^4. There is no top-degree cancellation. An independent diagonal coefficient certificate modulo

$1,000,003$ gives the nonzero final leading residue 580506, independent of every fixed coefficient leaf. Therefore

$$\deg P_{16} = 9\,465\,217\,590\,332\,031\,250\,000\,010\,008.$$

The delivered circuit is `data/signed16.json`. Its selected evaluator uses 1590 live binary gates. The large digit polynomial is evaluated using the complete homogeneous polynomial h_4. Set $t_j = B^{5^j}$, $v_j = B^{5^{31}-5^j}$, and $t = t_{31}$. Then

$$e_*(B) = t\left[z_* h_4(t, v_0, \dots, v_{31}) + \sum_{|\alpha|\le 4} c_\alpha t^{4-|\alpha|} \prod_j v_j^{\alpha_j}\right].$$

The exponent of a summand is exactly $5\cdot 5^{31} - \sum_j \alpha_j 5^j$. Initialize $h_0 = 1$ and $h_1 = h_2 = h_3 = h_4 = 0$. The ascending recurrence $h_r \leftarrow h_r + x h_{r-1}$, for $r = 1, 2, 3, 4$, evaluates the first term; multivariate Horner arithmetic evaluates the sparse correction. Shared fifth powers give t_j, while $v_j = t_j^4 v_{j+1}$. Finally

$$\lambda = \prod_{j=0}^{31} (1 + t_j + t_j^2 + t_j^3 + t_j^4)(1+q)(1+q^2).$$

These identities are polynomial identities; the output circuit has no division.

11 An identical twelve-variable polynomial in 108 operations

This improvement preserves the exact fixed polynomial of package 30. Its variables are $J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}$. The four constants are the same complete tuple as before. Define

$$\begin{gathered}
q = \beta J + 1,\ C_w = q - F - \alpha,\ X = wq,\ Y = sq^3,\ E = XY,\\
k = \eta + \zeta,\ c = kY + \eta,\ a = E + Y,\ H = 4a + 3,\ \Delta = a^2 + H,\\
\ell = EkY, \quad A = 1 + \ell(\ell + k), \quad G = 1 + \Delta c^2, \quad b_0 = X + ac,\\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_k = k - hE - R, \qquad N_t = (K_0 + w)C_w + (q - F) - z(q-1),\\
I = ic^2, \quad D = 1 + \Delta I^2, \quad Q_0 = \Delta(D-1), \quad d = Q_0 - 1,\\
v = cT, \quad b = c + RD, \quad M = k(I^2 + T^2), \quad L = HM + 1, \quad W = b_0 + Ht.
\end{gathered}$$

A complete scalar evaluator for the norm $\mathcal{N}$ is

$$\begin{aligned}
H_2 &= H^2, & H_Q &= H_2 Q_0, & H_P &= H_Q D v^2, & H_U &= H_Q b^2,\\
n &= (HM)^2 A, & d_n &= dn, & H_e &= H_P + H_U - H_2, & j_0 &= 4H_P H_U,\\
W_2 &= W^2, & L_2 G &= L^2 G, & \mathrm{Tr} &= 2d(W_2 + L_2 G),\\
K_r &= d(W_2 - L_2 G), & K_n &= K_r^2, & A_A &= 2(d_n + H_e),\\
C_c &= (d_n - H_e)^2 - j_0, & a_t &= \mathrm{Tr} - A_A, & b_t &= C_c - K_n,\\
c_f &= 16 d_n j_0, & v_r &= a_t(a_t \mathrm{Tr} + 2b_t) - c_f,\\
u_r &= b_t^2 - a_t^2 K_n.
\end{aligned}$$

The final norm is

$$\mathcal{N} = u_r(u_r + \mathrm{Tr} v_r) + K_n v_r^2.$$

The equation is

$$\boxed{P_{12} = N_k N_t(1 + \mathcal{N}) - 1 = 0.}$$

It has degree 1081 and uses 108 operations. The system $N_k = 1$, $N_t = 1$, $\mathcal{N} = 0$ uses 104 operations, with maximum residual degree 1072. Replacing the transport variable by $z_o = z + 1$ gives the comparison $(K_0 + w)C_w = F + z(q - 1)$ and saves one more operation. The positive-domain justification is the transport argument already given.

Proposition 11.1. *The 108-operation circuit is identical as an integer polynomial to the 111-operation source circuit. Its consistency equivalence is unchanged.*

Proof. There are precisely three algebraic savings. First share H^2Q_0 before forming H_P, H_U, saving one multiplication. Next compute $[d(W^2 - L^2G)]^2$ instead of $d^2(W^2 - L^2G)^2$, saving one multiplication. Finally Horner-evaluate $u_r^2 + \mathrm{Tr}u_rv_r + K_nv_r^2$ as in the display, saving one gate.

For an independent norm derivation, let $P_a = Q_0Dv^2$, $U_a = Q_0b^2$, $e_a = P_a + U_a - 1$, and $B_a = 2Q_0vb$. Then $\mathcal{N}$ is the eight-factor product

$$\prod_{u^2=G,\ \tau^2=A,\ f^2=D} [d(W - Lu - HM\tau)^2 - H^2e_a + H^2B_af],$$

where each root has both signs. Eliminating τ, f first gives a quartic in $x = d(W \pm L\sqrt{G})^2$ of the form $(x^2 - A_Ax + C_c)^2 - c_fx$. The two x values have sum Tr and product K_n. Reducing modulo $x^2 - \mathrm{Tr}x + K_n$ gives $u_r + v_rx$, whose norm is the displayed $\mathcal{N}$. All identities hold over polynomial rings, including at repeated or zero roots. The source's full positive-domain soundness proof is preserved and applies to the identical polynomial. No new compiler binding is required. □

12 A smaller complete coefficient tuple

The arithmetic constructions and the coefficient compiler can be improved independently. The present coefficient construction retains package 27's ordered Wang943 tile language and its full native/Pell interface, but reduces the base from $2^{1953125}$ to

$$b = 125, \qquad L = 8101, \qquad d = bL = 1012625, \qquad B = 2^d.$$

Thus the base exponent decreases by 48.1536%. The four actual integer ports occupy 506644 bytes in minimal unsigned binary form, supplied in 10502 bytes of deterministic gzip files. The largest port has 1015251 bits. These are coefficient improvements; no variable, degree, or operation claim follows merely from changing the tuple. The main signed-sixteen instance in this report retains its original fixed tuple.

12.1 Finite recipe

Put $\rho = 2^{125}$ and $W_j = 1024 \cdot 4^j$ for $0 \le j < 44$. Use the same 943 ordered tiles x_{it}, with ports $t = 0, 1, 2, 3$ denoting west, north, east, south. The ten-bit code of an edge label x is $x + 512$. The four copy exponent maps are

$$e_0(j) = 1024 + 2j, \quad e_1(j) = 1004 + 2j, \quad e_2(j) = 984 + 2j, \quad e_3(j) = 964 + 2j \quad (0 \le j < 10).$$

Let E consist of selectors $0, \dots, 942$, these forty copy positions, the dummies $1065, 1066$, and anchors $1110, 1112, 1111, 2225$, in that clause-weight order. In particular $|E| = 989$. Define

$$c_i = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512)W_{10t+j} + \sum_{j=40}^{43} W_j \quad (0 \le i < 943).$$

At a copy position $e_t(j)$ set $c_{e_t(j)} = W_{10t+j}$; at the four anchors assign $W_{40}, W_{41}, W_{42}, W_{43}$ respectively; at both dummies assign zero. Finally put

$$\begin{aligned}
\mu &= 1022 + \sum_{j=0}^{43} W_j, \\
D_C &= \rho + \rho^{40} + \rho^{61} + \rho^{1113} + \sum_{e\in E} c_e(\rho^{3341-e} + \rho^{4459-e}), \\
D_R &= \rho^{21}, \\
M_F^{\mathrm{nat}} &= 4 + \mu(\rho^{3341} + \rho^{4459}) + \sum_{j=0}^{9}(\rho^{1004+2j} + \rho^{1045+2j}) \\
&\qquad + \rho^{1111} + \rho^{2225} + \sum_{z=6685}^{7544} \rho^z, \\
M_C &= B - 1 - \sum_{\substack{e\in E \\ e\neq 1}} \rho^e - 2\rho^{1065}.
\end{aligned}$$

The complete coefficient tuple is

$$\boxed{(B-1,\ D_C + BD_R,\ M_C,\ M_F^{\mathrm{nat}} + B - 1).}$$

The fourth port is already shifted. All four ports must be replaced together. Every coefficient is fixed by bounded finite arithmetic, independently of the existence of a ZFC refutation.

12.2 Decoding and geometry

The mixed-radix clause decoder is unchanged. Before decoding, the low ten bits count at most 943 selected tiles; all other contributions are multiples of 1024. Mask 1022 forces that count to be zero or one. After this extraction each radix-four digit is a sum of at most two Boolean bits, and its masked low bit forces the required equality. Thus an occupied cell contains one tile, its exact forty edge bits, and four anchors equal to one.

The new geometry is summarized by the complete native-basis identities below. Each row specifies the only nonzero contribution from the indicated source; an empty entry means no contribution.

Test target	D_C predecessor	D_R predecessor	Temporal predecessor
$1004 + 2j$	$964 + 2j$		$1004 + 2j$
$1045 + 2j$	$984 + 2j$	$1024 + 2j$	
1111	1110		1111
2225	1112		2225

Here $0 \le j < 10$. For example, at the north target the unwanted bare predecessors under shifts 61 and 1 occupy the odd intervals $943, \dots, 961$ and $1003, \dots, 1021$; both miss E. The other rows follow by the same displayed finite arithmetic.

Write

$$S = [0, 942] \cup \{964, 966, \dots, 1042\} \cup \{1065, 1066, 1110, 1111, 1112\}, \qquad E = S \cup \{2225\}.$$

The span of S is 1112. Clause contamination at a target t would require $3341 - t$ or $4459 - t$ in $E - E$. All these differences exceed 2225 except $3341 - 2225 = 1116$; that exception is absent because $2225 - 1116 = 1109 \notin E$. At a clause target all bare and spatial predecessors lie above E: the smallest is $3341 - 1113 = 2228$.

The band separation 1118 exceeds the span of S, and $2225 - 1118 = 1107 \notin E$. Therefore $1118 \notin E - E$. Since $L > 2 \max E$, the two possible rotation-hit sets $(3341 - E) \bmod L$ and

$(4459 - E) \bmod L$ are disjoint. Every binary rotation leaves a clean clause band. Likewise the unique pair of elements of E with difference 1114 is $(1111, 2225)$, whence

$$\{1111 - e : e \in E\} \bmod L \ \cap \ \{2225 - e : e \in E\} \bmod L = \{0\}.$$

The source two-anchor parity argument first excludes nonzero offsets within a radix digit, including the one possible upper-dummy spill, and this intersection then forces whole-cell alignment. The edge and occupancy conditions give exactly the same Wang matching contract as before.

The conservative source carry bound is also unchanged. With

$$\mathcal{M} = 991 \left(2 \sum_{e \in E} c_e + 6 \right)$$

one has $4\mathcal{M} + 8 < 2^{125}$ and $2\mu + 4 < 2^{125}$; their relevant bit lengths are respectively 120 and at most 99. These estimates hold before selector extraction. The maximal unshifted convolution exponent is $4459 + 2225 = 6684$. The 860 mask fields at $6685, \ldots, 7544$ are therefore identically zero for every aligned intended cell, including every safe dummy adjustment. They fit below $L = 8101$ and restore the exact population balance

$$\mathrm{pc}(M_F^{\mathrm{nat}}) = 989, \qquad \mathrm{pc}(M_C) = d - 989, \qquad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8.$$

12.3 The rank-control extension

The source chooses a power-of-five cell length to tune the actual packed rank. The next argument isolates the necessary arithmetic condition and allows $L = 8101$.

Boolean subset lemma. Let $N = 5^n$, $M = dN$, and $T = N/5 > 5d$. Every residue modulo M is a Boolean subset sum of the distinct physical weights

$$\{B^{4j} : 0 \le j < T\} \ \cup \ \{B\}.$$

Indeed $2^{125} \equiv 90 \pmod{8101}$ and $90^2 \equiv -1 \pmod{8101}$, so $B^4 \equiv 1 \pmod{8101}$. Moreover

$$v_5(B^4 - 1) = v_5(2^{4d} - 1) = v_5(2^4 - 1) + v_5(d) = 4.$$

The Chinese remainder theorem implies that B^4 has exact order T modulo M. Its powers are consequently exactly the T residues $1 + 5dt$, with $t \bmod T$.

For a target z, choose $\epsilon \in \{0, 1\}$ so that $z - \epsilon B \not\equiv 0 \pmod 5$. Choose $1 \le k \le 5d$ congruent to $z - \epsilon B$ modulo $5d$. Then $k < T$ and k is invertible modulo T. Set

$$u = \frac{z - \epsilon B - k}{5d} \pmod T, \qquad r = k^{-1} \left(u - \frac{k(k-1)}{2} \right) \pmod T.$$

The k distinct cyclic residues $r, r + 1, \ldots, r + k - 1$ sum to u modulo T. The corresponding distinct powers B^{4j}, together with the optional B, sum to z modulo M. This uses Boolean choices, with no repeated physical cell or negative multiplicity.

For the canonical helical witness choose independently sufficiently large powers of five w, h, with $N = wh$ and $w < N/5$. The selected source positions are at most $4N/5 - 4$; their spatial and temporal destinations remain below N. Thus the inherited exact nonwrapping rank-difference formula applies. A dummy change at cell j changes the actual rank by $-\Gamma B^j$, where

$$\Gamma = \rho^{1065}(q^2 - 1)[1 + q(D_C + BD_R + B^w)], \qquad q = B^N.$$

To see that Γ is a unit modulo dN, work modulo 5 and 8101. In both moduli $\rho^2 = -1$, and $L, N, w \equiv 1 \pmod 4$, so $B = q = B^w = \rho$. The paired clause terms cancel since $1118 \equiv 2 \pmod 4$. The bare terms give $D_C = 1 + 3\rho$ and $D_R = \rho$. It follows that

$$q^2 - 1 = -2, \qquad 1 + q(D_C + BD_R + B^w) = -3.$$

As $\gcd(8101, 30) = 1$, both factors are units; the factor ρ^{1065} is a unit as well. The Boolean lemma therefore tunes the actual rank to $R \equiv dw \pmod{dN}$. No primality assumption on 8101 is required.

12.4 Transfer of the full interface

The geometry proves soundness by the same clean-band decoder and Wang-plane argument. Completeness begins with a genuine finite blank halt, constructs its padded helical word, and tunes that word's actual rank as above. Every dummy change preserves the selector, edge, anchor, mask, and added-zero-field conditions. The source positive-slack and half-binomial/Pell constructions then produce fresh positive witnesses at this actual rank.

For clarity, the extra promise used by signed constructions survives as well. Put $t = dN$, $q = 2^t$, and use the source packed objects

$$S_0 = (C - 1) + qF, \qquad T_0 = (M_C J + 1) + q(M_F^{\mathrm{nat}} J - 1), \qquad M_0 = q^2.$$

They satisfy $S_0 \,\&\, T_0 = 0$, $0 \le S_0, T_0 < M_0$, and $\mathrm{pc}(T_0) = t + 2$. The exact identity

$$R = (M_0 - S_0)(M_0 - 1) + T_0 = (M_0 - 1 - S_0)M_0 + (S_0 + T_0)$$

gives $\mathrm{pc}(R) = 3t + 2$, while $R \equiv 3 \pmod 4$. Since t is odd, the source witness $X = 2^R$ has square scale $X/q = 2^{R-t}$. Its ordinary Pell construction supplies the same odd positive multiplier. Finally the authentic half-binomial valuation gives $v_2(Y) = \mathrm{pc}(R) - 2 = 3t$; from $Y = sq^3$ it follows that s is positive and odd. Thus this tuple supplies the complete source interface, including the odd-scale promise, and can be used by a compatible arithmetic skeleton after explicitly rebinding all four ports.

The accompanying programs check all 69230 native-basis identities, materialized populations and residues, and an independently reconstructed coefficient tuple. They also construct 1029 Boolean subsets in a small composite analogue and 32 subsets for the actual modulus $d \cdot 5^{11} = 49444580078125$. These finite checks support the finite recipe; the universal mathematical argument is the proof above, composed with the identified inherited compiler and K215 calibration. This is neither a consistency proof nor a new end-to-end proof-assistant certificate.

13 Verification, reproducibility, and limitations

The archive is self-contained for its stated finite checks. All release scripts use Python's standard library; a TeX installation is needed only to rebuild this PDF. Run from the extracted directory:

```
python3 reproduce.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The first command rebuilds the circuits and coefficient recipes and runs the checks. It does not search for a ZFC refutation or a Diophantine solution.

The exact checks include all twenty-two affine residual identities, liveness and operation counts, an independent expansion of the shifted quartic and all 391 weighted coefficient recipes, the strict index bounds, and the three-root norm identity as equality of sparse polynomials. The concrete native payloads are authenticated after decompression. The signed circuit is also compared with direct displayed formulas at actual-coefficient residues, with literal 58,905-term digit-sum evaluations. Constructed Pell residue examples and intersectivity checks test transcription at finite values. The twelve-variable circuit is compared both with the predecessor and with an independent eight-conjugate evaluator.

Finite checks supplement the universal proofs. They do not establish consistency, prove a nonhalting computation by testing initial steps, or replace the imported unbounded native

compiler theorem. No new end-to-end Lean or Rocq certificate is claimed. In particular the paper states a provable equivalence; it does not assert that ZFC proves either side. Under the usual consistency hypothesis, the nonexistence assertion is unprovable in ZFC by the second incompleteness theorem. No stronger two-sided independence claim is needed here.

The source provenance separates three contributions: the inherited K215, Pell, compiler, and Jones mathematics; exact refactorings of existing polynomials; and the new guard-removal and parity-weight arguments. The supplied sixteen-variable circuit is a concrete new specialization; we make no assertion of priority for general Pell periodicity, radical norm merging, or intersective polynomials.

14 Further research questions

1. Can the current eight-coordinate Jones ratio predicate lose another guard without imposing a restriction on possible bases? In particular, replacing the base slack by a square is not justified: the recovered base must also be a power of two, and cofinality of squares alone does not solve that simultaneous condition.

2. Can a Pell parameter already present in the predicate supply the residue variable v in Lemma 8.2? A proof would need residue-class surjectivity compatible with the full ratio predicate; arbitrary cofinal Pell solutions are insufficient.

3. Can one remove a register from the 31-variable quadratic system, giving a normalized quartic in thirty variables? This would lower the Jones radix exponent by a factor of five, and is more valuable than a small local gate saving. Positivity of any restored coordinate must be proved.

4. Can the available ten-integer representation theorem be instantiated with a coefficient recipe and arithmetic DAG for this exact ZFC refutation predicate? That would address the gap between a general existence theorem and the concrete sixteen-variable release.

5. Can the fixed-coefficient choice be optimized jointly with the quartic index rather than measuring the native radix alone? The final y_* size depends on the shifted quartic coefficient bounds and on ν.

6. Can the positivity-normalization theorem and the parity-sensitive three-root finalizer be formalized as reusable Lean or Rocq lemmas, then composed with the existing literal Jones and K215 endpoints? This would reduce the gap between the new mathematical proof and an end-to-end machine-checked theorem.

7. Can the native compiler receive a standalone kernel-checked certificate covering its synchronization, carry bounds, and rank-control argument? Coefficient hashes and finite geometry checks alone do not certify that unbounded theorem.

8. Is a single equation below 67 operations attainable under the same binary-gate convention, or a smaller low-degree system without increasing the witness count? The present results do not settle either question.

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